

Creditor quidam ita convenit cum Debitore suo, ut singulis momentis pars proportionalis usurae annuae annumeretur sorti, quantum ipsi finito anno debeatur?

Esto sortis a , usura annua b : Jam si usura sorti non adjiceretur, finito anno daretur $a + b$: si singulis semestribus usurae pars proportionalis adjicienda sorti, elapso primo semestri deberetur $a + \frac{1}{2}b$, hinc pro secundo semestri dicendum: sortis a , uno semestri affert $a + \frac{1}{2}b$: quid affert $a + \frac{1}{2}b$? Quia ita debetur finito anno $aa + ab + \frac{1}{4}bb$, hoc est, $\frac{a(a+b)}{2}$: Deinde si singulis quadrimestribus usurae pars conveniens annumeranda sorti, debetur exacto primo quadrimestri $a + \frac{1}{3}b$, pro secundo fiat: ut a ad $a + \frac{1}{3}b$, ita $a + \frac{1}{3}b$ ad $\frac{a(a+b)}{3}$, pro tertio vero, ut a ad $a + \frac{1}{3}b$, ita $\frac{a(a+b)}{3}$ ad $\frac{a(a+b)}{3}$, debetur ergo finito anno $\frac{a(a+b)}{3}$. Porro si singulis trimestribus usura fuisset adjicienda sorti, exacto primo trimestri deberetur $a + \frac{1}{4}b$, exacto secundo $\frac{a(a+b)}{4}$, tertio $\frac{a(a+b)}{4}$, & tandem quarto, hoc est, finito anno $\frac{a(a+b)}{4}$. Pari ratione invenietur quod si singulis quintis aut sextis partibus anni usura adjiciatur sorti, id quod debetur post annum fore $\frac{a(a+b)}{5}$, vel $\frac{a(a+b)}{6}$. unde ~~concludenda~~ atque indefinite si singulis n partibus anni usura sorti apponenda, post annum debitum in $\frac{n(a+b)}{n-1}$, unde si n sit pars anni infinitesima seu momentum, problema est redactum videmus, ut pro eo solvendo opus tantum sit, potestatem infinitesimam ex sorte & parte infinitesima usurae annuae dividere per potestatem unam minorem infinitesima ipsius sortis: quod sic fiet: scribantur ordine omnes Potestates, primo $a + b$; deinde quad. ex $a + \frac{1}{2}b$; postmodum cub. ex $a + \frac{1}{3}b$; Biquad. ex $a + \frac{1}{4}b$ etc. hanc rationem & dividantur per $1, a, aa, a^3$ etc. hac ratione:
 Divisa per $1) \quad 1a \quad + \quad 1b$
 $a) \quad 1aa \quad + 2a \times \frac{1}{2}b \quad + 1 \times \frac{1}{2}b$
 $aa) \quad 1a^3 \quad + 3aa \times \frac{1}{3}b \quad + 3a \times \frac{1}{3}b \quad + 1 \times \frac{1}{3}b$
 $a^3) \quad 1a^4 \quad + 4aa \times \frac{1}{4}b \quad + 6aa \times \frac{1}{4}b \quad + 4a \times \frac{1}{4}b \quad + 1 \times \frac{1}{4}b$
 $a^4) \quad 1a^5 \quad + 5aa \times \frac{1}{5}b \quad + 10aa \times \frac{1}{5}b \quad + 10aa \times \frac{1}{5}b \quad + 5a \times \frac{1}{5}b \quad + 1 \times \frac{1}{5}b$
 quorum ultimi numero potestatum existente n , generaliter inveniuntur

$1, n, \frac{n(n-1)}{2}, \frac{n(n-1)(n-2)}{2 \times 3}, \frac{n(n-1)(n-2)(n-3)}{2 \times 3 \times 4}, \frac{n(n-1)(n-2)(n-3)(n-4)}{2 \times 3 \times 4 \times 5}$ etc.
 Ubi considerandum, coefficientes primo terminorum esse unitates, secundum numeros naturales, tertium trigonales, quartum pyramidalis,
 sicut $\frac{n(n-1)}{2}a + \frac{1}{2}b$
 hoc est, quia $a^{(n)}, a^{(n-1)}, a^{(n-2)}, a^{(n-3)}$ etc. divisa per $a^{(n)}$ producant $a, 1, \frac{1}{a}, \frac{1}{a^2}$ etc. in istam:
 $a + n \times \frac{1}{n}b + \frac{n(n-1)}{2a} \times \frac{1}{n}bb + \frac{n(n-1)(n-2)}{2 \times 3aa} \times \frac{1}{n}b^3 + \frac{n(n-1)(n-2)(n-3)}{2 \times 3 \times 4a^3} \times \frac{1}{n}b^4$ etc. five in hanc:
 $a + \frac{n}{n}b + \frac{n(n-1)}{2ann}bb + \frac{n(n-1)(n-2)}{2 \times 3aan^3}b^3 + \frac{n(n-1)(n-2)(n-3)}{2 \times 3 \times 4a^3n^4}b^4$ etc. Jam quoniam n existente ∞ , tantundem valent atque n , Equivalebit ista series huius
 $a + \frac{n}{n}b + \frac{nnbb}{2ann} + \frac{n^3b^3}{2 \times 3aan^3} + \frac{n^4b^4}{2 \times 3 \times 4a^3n^4}$ hoc est, huius $a + b + \frac{bb}{2a} + \frac{b^3}{2 \times 3aa} + \frac{b^4}{2 \times 3 \times 4a^3}$ etc.

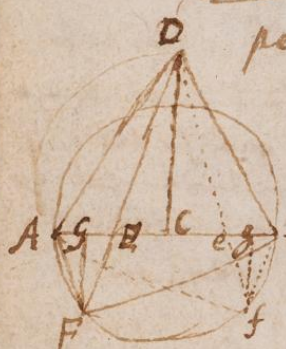
vel etiam quia $\frac{n \times n - 1}{n^2}, \frac{n \times n - 1 \times n - 2}{n^3}, \frac{n \times n - 1 \times n - 2 \times n - 3}{n^4} = \frac{n^2 - n}{n^2}, \frac{n^3 - 3n^2 + 2n}{n^3}, \frac{n^4 - 6n^3 + 11n^2 - 6n}{n^4} =$
 $1 - \frac{1}{n}, 1 - \frac{2}{n} + \frac{2}{n^2}, 1 - \frac{6}{n} + \frac{11}{n^2} - \frac{6}{n^3}$, etc: & vero per $n = \infty$ omnes hae fractiones adjectae unita-
 tibus evanescant, fit ut inde eadem resultet series, $a + b + \frac{bb}{2a} + \frac{b^3}{2 \times 3aa} + \frac{b^4}{2 \times 3 \times 4a^3}$ etc: quae
 cujus summa minor est $\frac{aa}{a-b}$, apparet quia $\frac{aa}{a-b} =$ series progressionis geometricae
 $a + b + \frac{bb}{a} + \frac{b^3}{aa} + \frac{b^4}{a^3}$ etc. quae altera illa minor est, AB. si supponatur $a = b$, fiet series
 ista: $2a + \frac{a}{2} + \frac{a}{2 \times 3} + \frac{a}{2 \times 3 \times 4} + \frac{a}{2 \times 3 \times 4 \times 5}$ etc: Repensio autem per subtractionem hujus seriei à se-
 ipso, quod $2a + \frac{a}{2} + \frac{2a}{2 \times 3} + \frac{3a}{2 \times 3 \times 4} + \frac{4a}{2 \times 3 \times 4 \times 5}$ etc: hoc est $2a + \frac{a}{2} + \frac{a}{3} + \frac{a}{2 \times 4} + \frac{a}{2 \times 3 \times 5} + \frac{a}{2 \times 3 \times 4 \times 6}$
 $= 3a$, adeoque $2a + \frac{a}{2} + \frac{a}{2 \times 3} + \frac{a}{2 \times 3 \times 4}$ etc $= 3a$

CLL.

Examen modi Renaldiniani inscribendi quiv polygona regularia in circulo,
 de promptu ex Lib. II. de Reptol. & Compos: Mathem: p. 367. (vid. Sturmii
 Mathesin euclideam p. 38.)

Modus hic est: Fiat triang: equil: ABD. Divisag diametris AB in tot partes aequales,
 quot laterum est figura inscribenda. Duabus earum permissis ab a' B usque A, duces
 per initium tertiae recta DF, & hinc recta FB, quam putat esse latus polygoni optati.
 Anal: sit secta diameter utique in E, e, ductaeq DEF, & FB & demissa in diametrum pe-

pendicularis FG, fiat CB = a, CE vel Ce = b, FB = x, unde AF = $\sqrt{4aa - x^2}$



AB. BF :: BF. BG. BA. AF :: AF. AG. $\frac{AG \times GB}{4aa} = GFq$
 $2a. x :: x. \frac{x}{2a}$ $\frac{2a \times x}{4aa} = \frac{x^2}{4aa}$
 $\frac{GB \times CB - CE}{4aa} = \frac{GEq}{4aa}$
 $\frac{x^2}{4aa} = \frac{GEq}{4aa}$
 $x^2 = 2a^2x \pm 2a^2bx + 4aabbx - 2ab - 2a^2bb \pm 2a^2b$
 $3aa + bb$

$x = \sqrt{2aa \pm 6a^2b \pm 2aabb \pm 3aa - 2bb}$ Jam si juxta regulam Renaldini abscondatur pro

Trigono $\frac{2a}{3}$ $\frac{BE}{3}$ $\frac{1}{3}a$ $\frac{1}{3}a$ $\frac{1}{3}a$ $\frac{1}{3}a$ $\frac{1}{3}a$ $\frac{1}{3}a$
 Quadrato $\frac{4a}{3}$ $\frac{4a}{3}$ $\frac{4a}{3}$ $\frac{4a}{3}$ $\frac{4a}{3}$ $\frac{4a}{3}$ $\frac{4a}{3}$ $\frac{4a}{3}$
 Pentagono $\frac{5a}{3}$ $\frac{5a}{3}$ $\frac{5a}{3}$ $\frac{5a}{3}$ $\frac{5a}{3}$ $\frac{5a}{3}$ $\frac{5a}{3}$ $\frac{5a}{3}$
 Hexagono $\frac{6a}{3}$ $\frac{6a}{3}$ $\frac{6a}{3}$ $\frac{6a}{3}$ $\frac{6a}{3}$ $\frac{6a}{3}$ $\frac{6a}{3}$ $\frac{6a}{3}$
 Heptagono $\frac{7a}{3}$ $\frac{7a}{3}$ $\frac{7a}{3}$ $\frac{7a}{3}$ $\frac{7a}{3}$ $\frac{7a}{3}$ $\frac{7a}{3}$ $\frac{7a}{3}$

sed reliqua duonon item, nam posito radio $a = 100000$, provenit pro
 Pentagono — 117492 justo valore minor q per tabulas sinuum reperitur 117557 — 13 min.
 Heptagono — 86918 justo valore major, quo reperitur — 86777 — 37 min.
 Octogono — 76030 — major — 76536 — 90 min.